

Syeda Sumayya Zahid

Software Engineer | Robotic And AI Enthusiast

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OBJECTIVE

Results-driven Software Engineer with comprehensive experience in full-stack web development, intelligent automation, and distributed robotic software systems (ROS 2). Proven expertise in architecting scalable applications, designing responsive user interfaces, and deploying optimized machine learning models for real-time decision-making. Strong foundation in software engineering methodologies, Agile frameworks, and the full Software Development Life Cycle (SDLC)

EXPERIENCE

AI Engineer | [Arbotrix](#)

Onsite | March 2026 – Present

- Developed and trained ML/DL models (CNNs, neural networks) for intelligent robotics and AI-driven production applications.
- Engineered ROS 2-based robotic software systems covering node communication, sensor integration, and autonomous operations.
- Designed and deployed full-stack web applications with RESTful APIs, responsive frontends, and scalable backend architectures.
- Integrated trained AI models into live robotic workflows enabling real-time decision-making and adaptive system behaviour.
- Performed end-to-end dataset preprocessing, model evaluation, hyperparameter tuning, and performance optimisation.
- Contributed to robotics software debugging, hardware-software interfacing, and system-level testing.
- Collaborated with cross-functional engineering teams to architect and deliver scalable AI-powered robotics solutions.
- Participated in R&D initiatives for next-generation AI, automation, and autonomous robotics technologies.

AI Model trainer | [TechKnock.tech](#)

Remote | March 2025 – Feb 2026

- Trained and optimised ML and DL models targeting high accuracy and low-latency real-time inference.
- Designed and managed end-to-end data pipelines: collection, cleaning, augmentation, and versioning for reproducible experiments.
- Conducted rigorous model evaluation and validation cycles to benchmark and improve reliability across diverse use cases.
- Collaborated with autonomous systems teams to integrate trained models into robotic and software application workflows.

PROJECTS

DODO Robot — Role: [AI and Robotic engineer](#)

- Integrated AI models into robotic workflows for automation and decision-making.
- Assisted in sensor handling, robot communication, and control system implementation.
- Participated in robotic system testing, debugging, and simulation.

- Contributed to AI model training and performance optimization for robotics applications.

Emotion Detection using EffectNet Dataset — *Role: AI and Model Trainer*

- Trained emotion recognition models using facial expression datasets.
- Applied preprocessing and augmentation techniques to improve model performance.
- Worked with computer vision techniques for emotion classification tasks.
- Evaluated model accuracy and optimized training workflows

AI Learn Hub - AI -powered Online platform Web App — *Role: Full stack developer*

- Architected a TypeScript full-stack application (React + Node.js + Firebase) with AI integration using Claude API for dynamic quiz generation from course content.
- Implemented Firebase Authentication with custom JWT tokens, role-based middleware (Student/Instructor/Admin), and secure session management.
- Built a RESTful API handling course CRUD, lesson management, enrollment flow, AI quiz generation/submission, and automated grading with weak-topic identification.
- Developed adaptive learning features including performance-based recommendations, progress tracking, and personalized result analytics.
- Designed the frontend with component-based architecture using React Router, Context API, and Tailwind CSS, ensuring responsive design across devices.

Tera X Autonomus Robot Hardware — *Role: Hardware engineer*

- Designed and assembled the Tera X robotic hardware platform including structural components and electronic systems.
- Integrated and wired actuators, cameras, proximity sensors, and embedded controllers into the physical robot body.
- Implemented hardware-software interfacing to enable communication between onboard sensors and ROS 2 software nodes.
- Conducted hardware testing, circuit debugging, and iterative prototyping to ensure mechanical and electrical reliability.
- Collaborated with the AI/software team to align hardware design with autonomous navigation and control system requirements.
- Performed simulation validation and real-world stress testing to verify system robustness under operational conditions.

Arbotrix - Company and Portfolio Website — *Role: Full stack developer*

- Designed and developed the official company website for Arbotrix.tech using Next.js, Node.js, and Tailwind CSS.
- Built a server-side rendered frontend with Next.js for fast load times, SEO optimisation, and dynamic routing.
- Implemented responsive UI components using Tailwind CSS ensuring a polished, professional experience across all devices.
- Developed backend logic and API routes with Node.js for dynamic content handling and efficient server-side processing.
- Managed full project version control using Git/GitHub including branching, commits, and production deployment pushes.
- Optimised performance, cross-browser compatibility, and mobile responsiveness aligned with the company's technical branding.

AI Rehman Welfare - Charity and Management Welfare Web App — *Role: Full stack developer*

- Developed a full-stack web application for a welfare organisation covering charity/donation management, volunteer coordination, and event management.
- Built an intuitive frontend interface enabling donors, volunteers, and admins to interact with the platform efficiently.
- Implemented backend systems for donation tracking, volunteer registration, and event scheduling with database integration.
- Designed an admin dashboard for managing records, monitoring activity, and generating operational reports.

- Applied Agile practices, Git version control, and iterative delivery to meet organisational requirements on schedule.

AI-Powered Emergency Response System — Role: *AI and System*

Development Final Year Project

- Developed an AI-powered emergency response application capable of detecting threats using voice emotion recognition and movement analysis.
- Integrated intelligent detection mechanisms for real-time emergency identification and alert generation.
- Worked on AI-based analysis techniques to improve response accuracy and public safety assistance.
- Contributed to system testing, documentation, and performance evaluation.

AI-Powered Skin Cancer Detection — Role: *AI/ML and Full stack developer*

- Developed a full-stack AI-powered skin cancer detection system using Next.js, FastAPI, TensorFlow, ResNet50, and SQLite.
- Trained and optimized a ResNet50 deep learning model on the HAM10000 dataset to classify 7 skin lesion types with high accuracy.
- Built secure REST APIs with JWT authentication, integrated OpenCV for image preprocessing/validation, and managed user data with SQLAlchemy and SQLite.
- Designed a responsive interface for real-time image analysis, prediction confidence scores, scan history, and medical recommendations using Next.js and Tailwind CSS.

RS Beauty Saloon — Role: *Full stack developer*

- Developed a responsive beauty salon management website using Next.js, React.js, Tailwind CSS, and Node.js with a modern, user-friendly interface.
- Built service, appointment booking, and contact management features, enabling customers to explore services and schedule appointments online.
- Designed an admin panel to manage salon services, customer inquiries, appointments, and business information efficiently
- Integrated secure backend APIs and database functionality for appointment management and data persistence.

EDUCATION

BS Software Engineering

Comsats University Islamabad, Abbottabad Campus | 2022 – 2026 | Grade : A+

FSc (Pre-Engineering)

Islamia College, Peshware | 2019 – 2021 | Grade: A+

TOOLS & SKILLS

AI & Machine Learning: Python, Machine Learning, Deep Learning, TensorFlow, Keras, Neural Networks, CNNs, Transfer Learning, Model Training & Optimisation, Hyperparameter Tuning, Data Preprocessing & Augmentation, AI Model Evaluation

Computer Vision: OpenCV, Object Detection, Image Classification, Facial Expression Recognition, Real-Time Video Processing

Robotics & Autonomous Systems: ROS 2, Autonomous Systems Development, Sensor Integration & Fusion, Hardware-Software Interfacing, Robot Simulation, Robotics Control Systems, AI for Robotics

Web Development: Next.js, Node.js, Tailwind CSS, HTML5, CSS3, JavaScript, RESTful API Development, Responsive Design, Database Integration, Server-Side Rendering

Software Engineering & QA: SDLC, Agile / Scrum Methodologies, API Testing, Manual Testing, Debugging, Requirement Analysis, Version Control (Git/GitHub)

Platforms & Tools: Google Colab, VS Code, Postman, Jira, GitHub

Communication & Collaboration: Slack, Microsoft Teams, Zoom, Confluence

Business & Productivity: Google Sheets, Microsoft Excel